[INCH-POUND]
A-A-51146B
September 5, 1996
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

FILES, ROTARY, CYLINDRICAL SHAPE, HIGH SPEED STEEL OR TUNGSTEN CARBIDE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers cylindrical shape high speed steel or tungsten carbide type rotary files.

1.2 Intended use. The rotary files covered by this CID are intended for general-purpose machining and deburring.

2. CLASSIFICATION

2.1 Classification. The rotary files shall be of the following types, styles, sizes and classes. The type, style, size and class to be furnished shall be as specified (see 7.2).

Type: Type I - High speed steel.
Type II - Tungsten carbide.

Style: Style A - Right hand spiral fluting.
Style B - Right hand spiral fluting with chip breaker.
Style C - Double cut fluting.
Style D - Diamond cut fluting.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3455

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Size: Unless otherwise specified (see 7.2), the cutter head length, cutter head diameter, and shank diameter of the type I and type II rotary files shall conform to tables I and II, respectively.

TABLE I. Type I, high speed steel rotary files, sizes and characteristics.

Size	Characteristics		
	Shank Diameter (inches)	Cutter Head Dimensions (inches)	
		Diameter	Head Length
1	1/8	1/8	5/8
2	1/8	1/4	1/2
3	1/8	1/4	5/8
4	1/8	3/16	5/8
5	1/4	1/8	1/2
6	1/4	1/8	3/4
7	1/4	3/16	1/2
8	1/4	3/16	3/4
9	1/4	1/4	5/8
10	1/4	1/4	1
11	1/4	5/16	3/4
12	1/4	5/16	1
13	1/4	5/16	1-1/2
14	1/4	3/8	1
15	1/4	3/8	1-1/2
16	1/4	1/2	1
17	1/4	1/2	1-1/2
18	1/4	5/8	1
19	1/4	3/4	1/2
20	1/4	3/4	3/4
21	1/4	7/8	1
22	1/4	1	1

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TABLE II. Type II, tungsten carbide rotary files, sizes and characteristics.

Size	Characteristics		
	Shank Diameter (inches)	Cutter Head Dimensions (inches)	
		Diameter	Head Length
1	1/8	1/16	1/4
2	1/8	1/16	3/8
3	1/8	3/32	7/16
4	1/8	1/8	9/16
5	1/8	1/4	1/4
6	1/8	1/4	1/2
7	1/4	1/8	1/2
8	1/4	1/8	5/8
9	1/4	5/32	5/8
10	1/4	3/16	5/8
11	1/4	1/4	5/8
12	1/4	5/16	3/4
13	1/4	3/8	3/4
14	1/4	7/16	1
15	1/4	1/2	1
16	1/4	5/8	1
17	1/4	3/4	1/2
18	1/4	3/4	3/4
19	1/4	3/4	1
20	1/4	1	1

Class: Unless otherwise specified (see 7.2), the spacing of teeth along the circumference of the cutter head shall be selected from Table III for the diameter of cutter head specified.

TABLE III. Number of teeth per circumference, type I rotary files.

Cutting Head Diameter (in.)	Teeth Per Circumference			
	Class A Coarse Cut	Class B Medium Cut	Class C Fine Cut	Class D Extra Coarse Cut
1/8	6 - 8	9 - 11	12 - 15	4 - 5
3/16	9 - 11	12 -14	15 - 20	4 - 8
1/4	10 - 13	14 - 16	17 - 21	4 - 9
5/16	11 - 16	17 - 20	21 - 26	4 - 10
3/8	12 - 17	18 - 22	23 - 30	5 - 11
7/16	13 - 18	19 - 24	25 - 31	6 - 12
1/2	14 - 20	21 - 26	27 - 35	7 - 13
5/8	17 - 22	23 - 29	30 - 40	9 - 16
3/4	18 - 24	25 - 32	33 - 44	10 - 17
7/8	19 - 25	26 - 33	34 - 45	14 - 18
1	20 - 27	28 - 41	42 - 55	15 - 19

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TABLE IV. Number of teeth per circumference, type II rotary files.

Cutting Head Diameter (in.)	Teeth Per Circumference			
	Class A Coarse Cut	Class B Medium Cut	Class C Fine Cut	Class D Extra Coarse Cut
1/8	8 - 11	12 - 15	16 - 20	4 - 7
3/16	10 - 12	13 - 16	17 - 24	4 - 9
1/4	12 - 15	16 - 20	21 - 26	4 - 11
5/16	13 - 16	17 - 22	23 - 30	4 - 12
3/8	13 -18	19 - 26	27 - 38	5 - 12
7/16	15 - 20	21 - 27	28 - 36	6 - 14
1/2	16 - 22	23 - 28	29 - 39	7 - 15
5/8	18 - 25	26 - 38	39 - 47	9 - 17
3/4	19 - 28	29 - 38	39 - 49	10 - 18
7/8	22 - 29	30 - 37	38 - 55	14 - 21
1	24 - 32	33 - 43	44 - 62	15 - 23

3. SALIENT CHARACTERISTICS

3.1 General requirements. The cutting heads of type I rotary files shall meet the minimum requirements for standard high-speed steel described in the annex of ASTM A600 - 92a when tested in accordance with ASTM A370. The cutting heads of type II rotary files shall be manufactured from tungsten carbide with a hardness of not less than 91.2 and not greater than 92.5 on the Rockwell "A" scale when tested in accordance with ASTM B294. The cutting heads of type II rotary files shall have a density of 14.6 to 15.1 grams per cubic centimeter when tested in accordance with ASTM B311. Type II rotary files shall have a transverse rupture strength of 220,000 pounds per square inch when tested in accordance with ASTM B406.

3.2 Fluting. The fluting of the rotary file shall extend along the length of the cylindrical cutter head in either right-hand spiral, right-hand spiral with chipbreaker, double cut, or diamond cut pattern, depending upon the style specified. Style A rotary files shall have right-hand spiral fluting consisting of a right-hand helix. The helix angle measured from the centerline of the cutter head shall be between 15° and 35° inclusive. Style B rotary files shall have right-hand spiral fluting with a left-hand spiral chipbreaker along the periphery of the cutting teeth. Style C rotary files shall have double cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals which results in a wedge or chisel form cutting tooth. The number of left-hand spirals shall be between 40 percent and 60 percent of the number of right-hand spirals. Style D rotary files shall have diamond cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals that results in a pyramidal form cutting tooth. The number of left-hand spirals shall be 70 percent to 90 percent of the number of right-hand spirals. When specified (see 7.2), straight end cut fluting shall extend from the centerline of the end of the cylinder to the circumference of the cylinder.

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4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered or environmentally preferable materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING

6.1 Packaging requirements. Preservation, packing, and marking shall be as specified (see 7.2).

7. NOTES

7.1 Source of documents.

7.1.1 Industry documents. Copies of industry standards referenced in this CID may be obtained from the following address:

American Society for Testing and Materials (ASTM)

ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM A600 - Standard Specification for Tool Steel High Speed

ASTM B294 - Standard Test Method for Hardness Testing of Cemented Carbides

ASTM B311 - Test Method for Density Determination for Powder Metallurgy (P/M) Materials Containing Less Than Two Percent

ASTM B406 - Standard Test Method for Transverse Rupture Strength of Cemented Carbides

Application for copies should be addressed to American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.

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b. Type, style, size, and class required (see 2.1).
c. Shank diameter, if different (see 2.1).
d. Cutter head diameter, if different (see 2.1).
e. Cutter head length, if different (see 2.1).
f. Cutter head teeth per inch (see 2.1).
g. Straight end cut fluting, if required (see 3.2).
h. Packaging (see 6.1).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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- Class: A - Coarse tooth spacing
  B - Medium tooth spacing
  C - Fine tooth spacing
  D - Extra coarse tooth spacing
- Size: See tables I and II
- Style: A - Right hand spiral fluting
  B - Right hand spiral fluting with chip breaker
  C - Double cut fluting
  D - Diamond cut fluting
- Type: 1 - Type I (High speed steel)
  2 - Type II (Tungsten carbide)
- CID number

MILITARY INTERESTS:

Custodians
Army - AL
Air Force - 99

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Air Force - 84

PREPARING ACTIVITY:

DLA - GS

(Project 3455-0155)

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